1_preprocess_exp1_sv_online_offline

Your Name

2026-08-05

0 prepare

library package

set path

1 read data - online & off line

rename

attention check

```
##
##
## ========================================
## 开始处理: online
## ========================================
##
## 【读取 sojump】
##   pre 数据行数: 113 ，唯一 id 数: 107
##   post 数据行数: 106 ，唯一 id 数: 105
##   subinf 数据行数: 117 ，唯一 id 数: 115
##
## 【sojump pre ID 修正】已应用  4  条规则
##
## 【sojump post ID 修正】已应用  3  条规则
##
## 【删除指定 ID】移除: SV221
##   修正后 pre 唯一 id 数: 108
##   修正后 post 唯一 id 数: 105
##
## 【去重后】
##   pre: 从 112 行减少到 108 行，唯一 id: 108
##   post: 从 105 行减少到 105 行，唯一 id: 105
##
## 【合并 sojump 后】
##   合并后行数: 105 ，唯一 id 数: 105
##   在 subinf 但不在 pre 中的 ID: SV221, SV642, SV643, SV644, SV645, SV646,
##   在 subinf 但不在 post 中的 ID: SV221, SV238, SV603, SV621, SV642, SV643, SV644, SV645, SV6
46,
##
## 【读取 Gorilla】
##   exp 数据行数: 51683
##
## 【Gorilla Response 修正】已应用  2  条规则
##
## 【提取 ID 映射】
##   映射表行数: 103 ，唯一 Participant_Public_ID 数: 103
##
## 【评分试次】
##   评分试次行数: 18858
##
## 【宽格式转换后】
##   宽格式行数: 6283 ，唯一 id 数: 103
##
## 【注意力检查】
##   未通过检查的被试 ID: SV203, SV212, SV240
##   准备移除 ID: SV203, SV212, SV240
##
## 【移除 attention 试次后】
##   剩余行数: 6000 ，唯一 id 数: 100
##
## 【合并 saq_sum】成功，saq_sum 已加入数据。
##
## 【处理完成】 online 最终被试数: 100
##
##
```

```
## ========================================
## 开始处理: offline
## ========================================
##
## 【读取 sojump】
##   pre 数据行数: 52 ，唯一 id 数: 52
##   post 数据行数: 52 ，唯一 id 数: 52
##   subinf 数据行数: 52 ，唯一 id 数: 52
##   修正后 pre 唯一 id 数: 52
##   修正后 post 唯一 id 数: 52
##
## 【去重后】
##   pre: 从 52 行减少到 52 行，唯一 id: 52
##   post: 从 52 行减少到 52 行，唯一 id: 52
##
## 【合并 sojump 后】
##   合并后行数: 52 ，唯一 id 数: 52
##
## 【读取 Gorilla】
##   exp 数据行数: 24280
##
## 【提取 ID 映射】
##   映射表行数: 31 ，唯一 Participant_Public_ID 数: 31
##
## 【评分试次】
##   评分试次行数: 9516
##
## 【宽格式转换后】
##   宽格式行数: 3172 ，唯一 id 数: 52
##
## 【GorillaID 映射后】
##   唯一 id 数: 52
##
## 【注意力检查】
##   所有被试均通过。
##
## 【移除 attention 试次后】
##   剩余行数: 3120 ，唯一 id 数: 52
##
## 【合并 saq_sum】成功，saq_sum 已加入数据。
##
## 【处理完成】 offline 最终被试数: 52
```

merge online & offline

combined data

```
## # A tibble: 6 × 13
##   id    id_public Trial_Number Spreadsheet_video valence arousal    sa rt_val
##   <chr> <chr>            <int> <chr>                <dbl>   <dbl> <dbl>  <dbl>
## 1 SV204 7y4e38v9             1 CT_POS_SI_075.mp4        7       2     7 10266.
## 2 SV204 7y4e38v9             2 CT_POS_SI_104.mp4        8       1     7  6292.
## 3 SV204 7y4e38v9             3 CT_POS_SI_051.mp4        8       1     8  5108.
## 4 SV204 7y4e38v9             4 CT_POS_SI_139.mp4        7       1     7  6299.
## 5 SV204 7y4e38v9             5 GC_POS_SI_004.mp4        7       1     6  8642.
## 6 SV204 7y4e38v9             6 GC_NEU_SI_135.mp4        5       2     3 27762.
## # ℹ 5 more variables: rt_aro <dbl>, rt_sa <dbl>, video_category <chr>,
## #   source <chr>, saq_sum <int>
```

```
##
## offline  online
##    3120    6000
```

2 statistic analysis

preprocess

function for analysis

2.1 anova

```
##
##
## ## ANOVA 与事后比较 ( Online )
```

```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## ℹ Please use `all_of()` or `any_of()` instead.
##   # Was:
##   data %>% select(dv)
##
##   # Now:
##   data %>% select(all_of(dv))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## ### ANOVA - 评分
##
##
## ### ANOVA - 反应时
##
##
## ### 事后比较 - 评分
##
##
## ### 事后比较 - 反应时
```

RT Pairwise (Online)

DV	group1	group2	statistic	df	p.adj
RT_Valence	POS	NEU	-3.569073	99	0.002000
RT_Valence	POS	NEG	-3.742459	99	0.000918
RT_Valence	NEU	NEG	-1.300980	99	0.588000
RT_Arousal	POS	NEU	-1.479880	99	0.426000
RT_Arousal	POS	NEG	-2.379208	99	0.058000
RT_Arousal	NEU	NEG	-1.674192	99	0.292000
RT_SA	POS	NEU	-2.746738	99	0.021000
RT_SA	POS	NEG	-2.593236	99	0.033000
RT_SA	NEU	NEG	-1.065180	99	0.867000

```
##
##
## ## ANOVA 与事后比较 ( Offline )
##
## ### ANOVA - 评分
##
##
## ### ANOVA - 反应时
##
##
## ### 事后比较 - 评分
##
##
## ### 事后比较 - 反应时
```

RT Pairwise (Offline)

DV	group1	group2	statistic	df	p.adj
RT_Valence	POS	NEU	-1.7672164	51	0.250
RT_Valence	POS	NEG	-2.0847115	51	0.126
RT_Valence	NEU	NEG	-0.4780099	51	1.000

DV	group1	group2	statistic	df	p.adj
RT_Arousal	POS	NEU	-0.8628992	51	1.000
RT_Arousal	POS	NEG	-0.0031583	51	1.000
RT_Arousal	NEU	NEG	1.2687161	51	0.630
RT_SA	POS	NEU	-0.9436610	51	1.000
RT_SA	POS	NEG	-1.0145852	51	0.945
RT_SA	NEU	NEG	-0.0419540	51	1.000

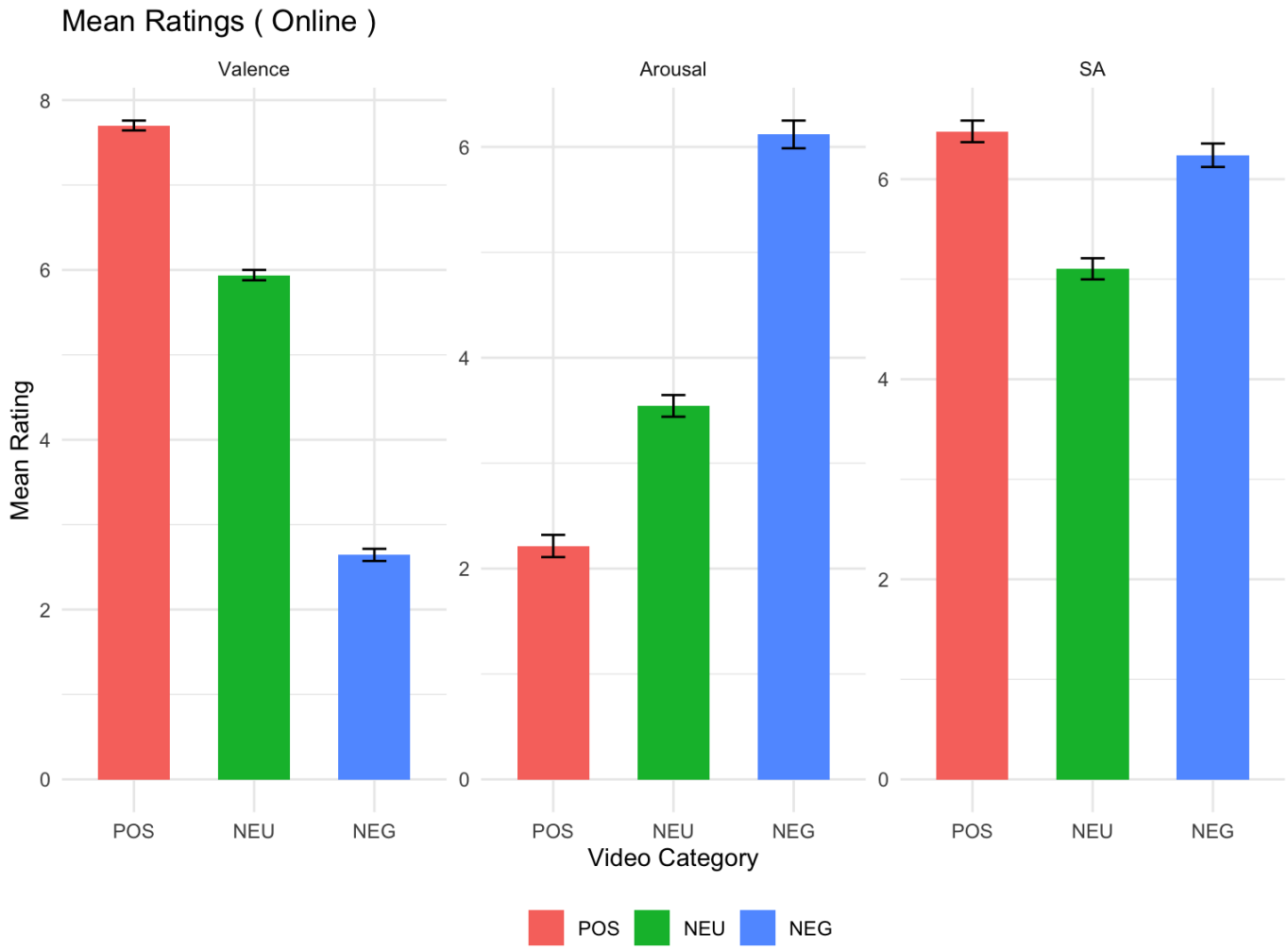
```
##
##
## ## ANOVA 与事后比较 ( Combined )
##
## ### ANOVA - 评分
##
##
## ### ANOVA - 反应时
##
##
## ### 事后比较 - 评分
##
##
## ### 事后比较 - 反应时
```

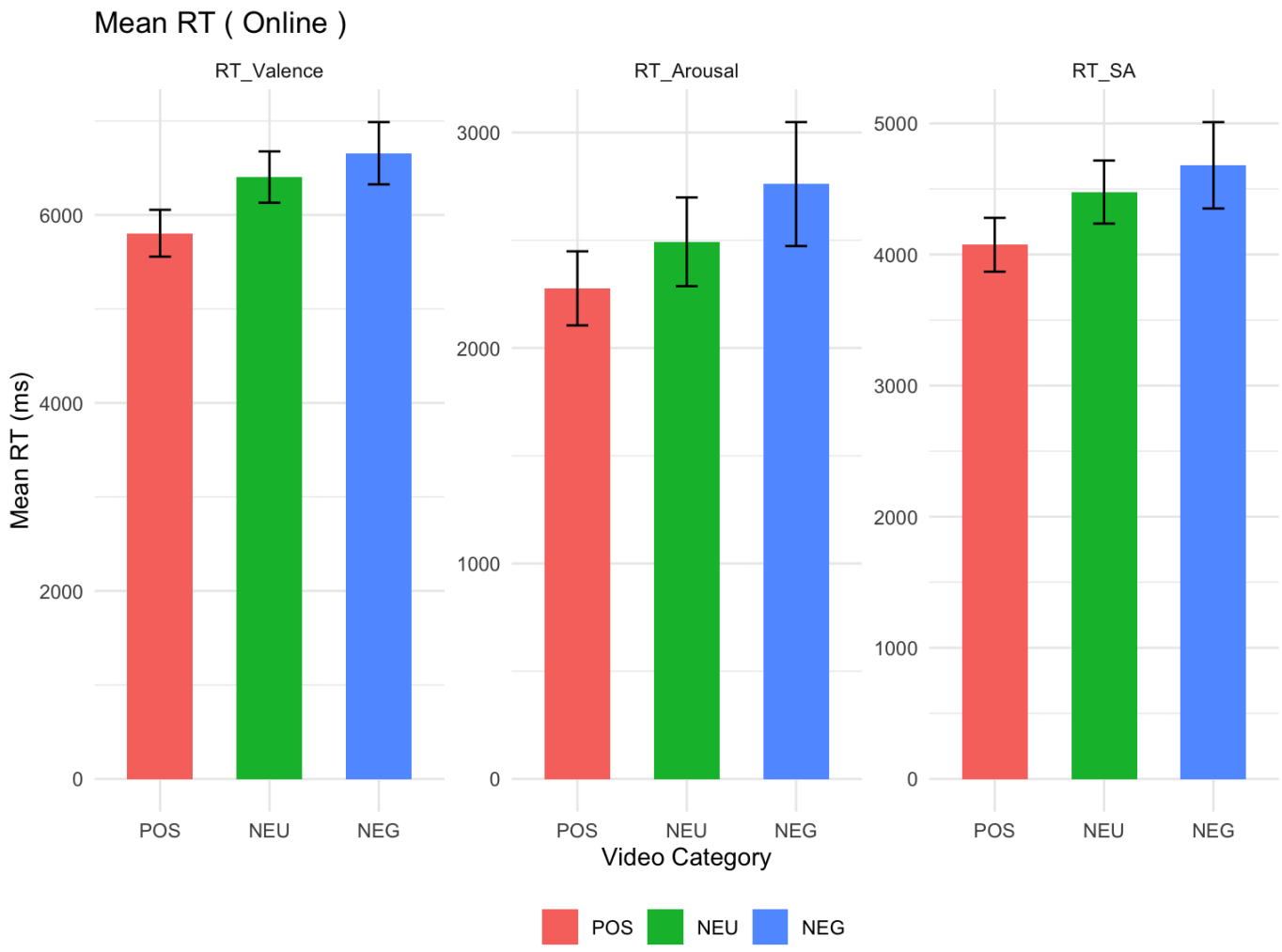
RT Pairwise (Combined)

DV	group1	group2	statistic	df	p.adj
RT_Valence	POS	NEU	-3.952686	151	0.000354
RT_Valence	POS	NEG	-4.171065	151	0.000153
RT_Valence	NEU	NEG	-1.386957	151	0.501000
RT_Arousal	POS	NEU	-1.714609	151	0.265000
RT_Arousal	POS	NEG	-2.277209	151	0.073000
RT_Arousal	NEU	NEG	-1.199002	151	0.696000
RT_SA	POS	NEU	-2.849945	151	0.015000
RT_SA	POS	NEG	-2.772976	151	0.019000
RT_SA	NEU	NEG	-1.010941	151	0.942000

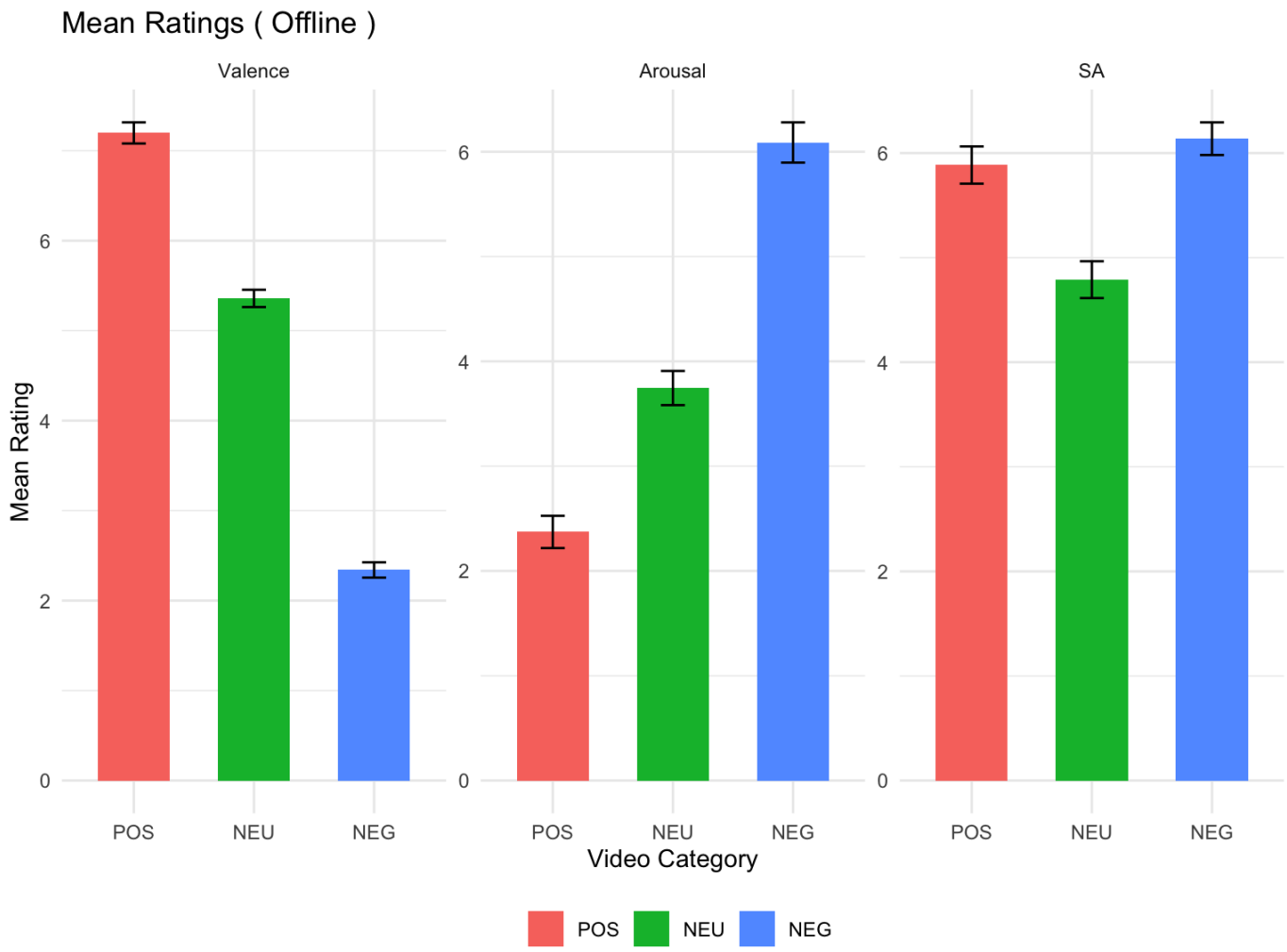
func:figure for anova

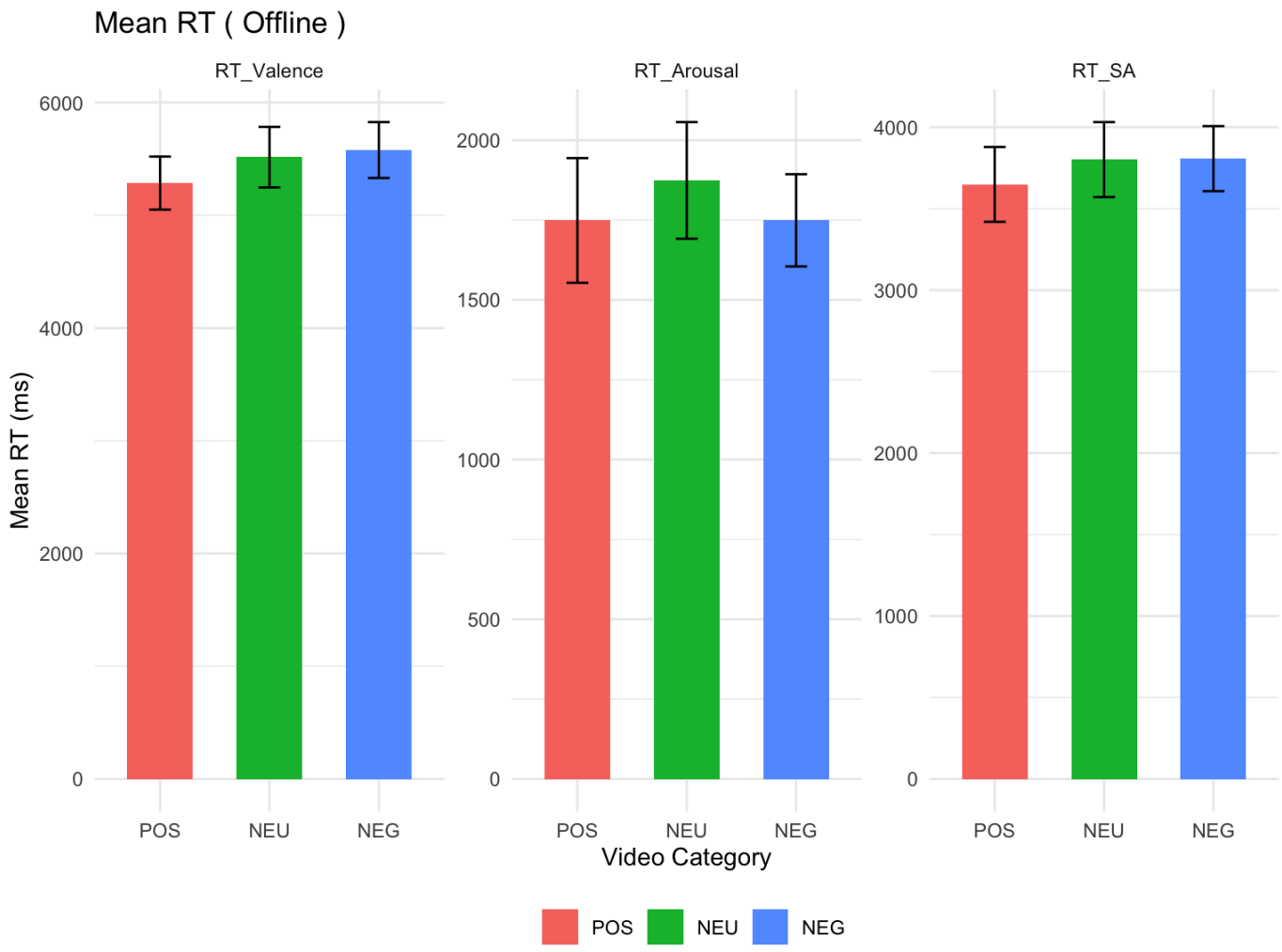
```
##
##
## ## 条形图 ( Online )
```


Mean Ratings (Online)
Valence
Arousal
SA
Mean Rating
0
2
4
6
8
POS
NEU
NEG
Video Category
POS
NEU
NEG

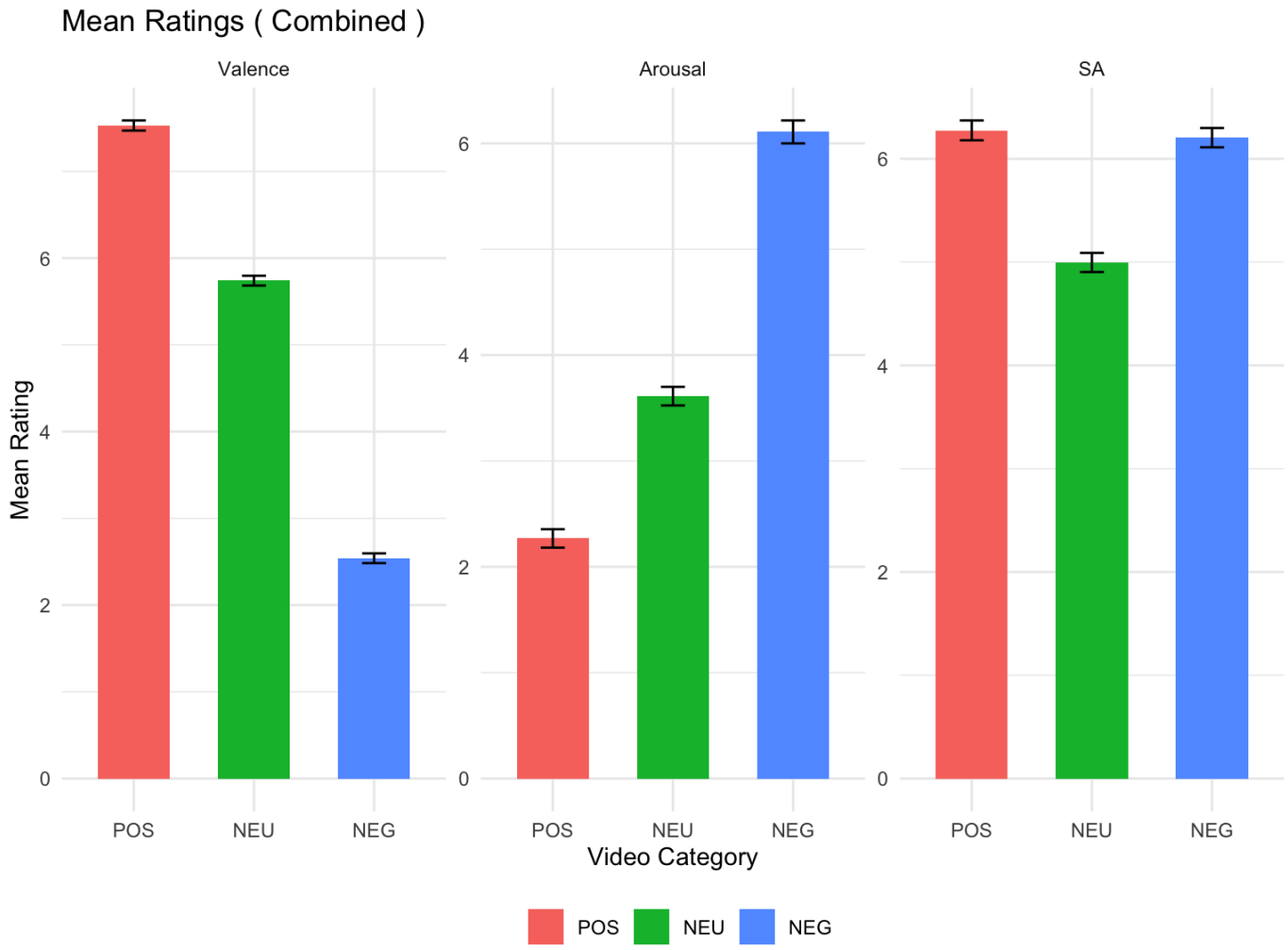

Mean RT (Online)
RT_Valence
RT_Arousal
RT_SA
Mean RT (ms)
0
1000
2000
3000
4000
5000
6000
POS
NEU
NEG
Video Category
POS
NEU
NEG

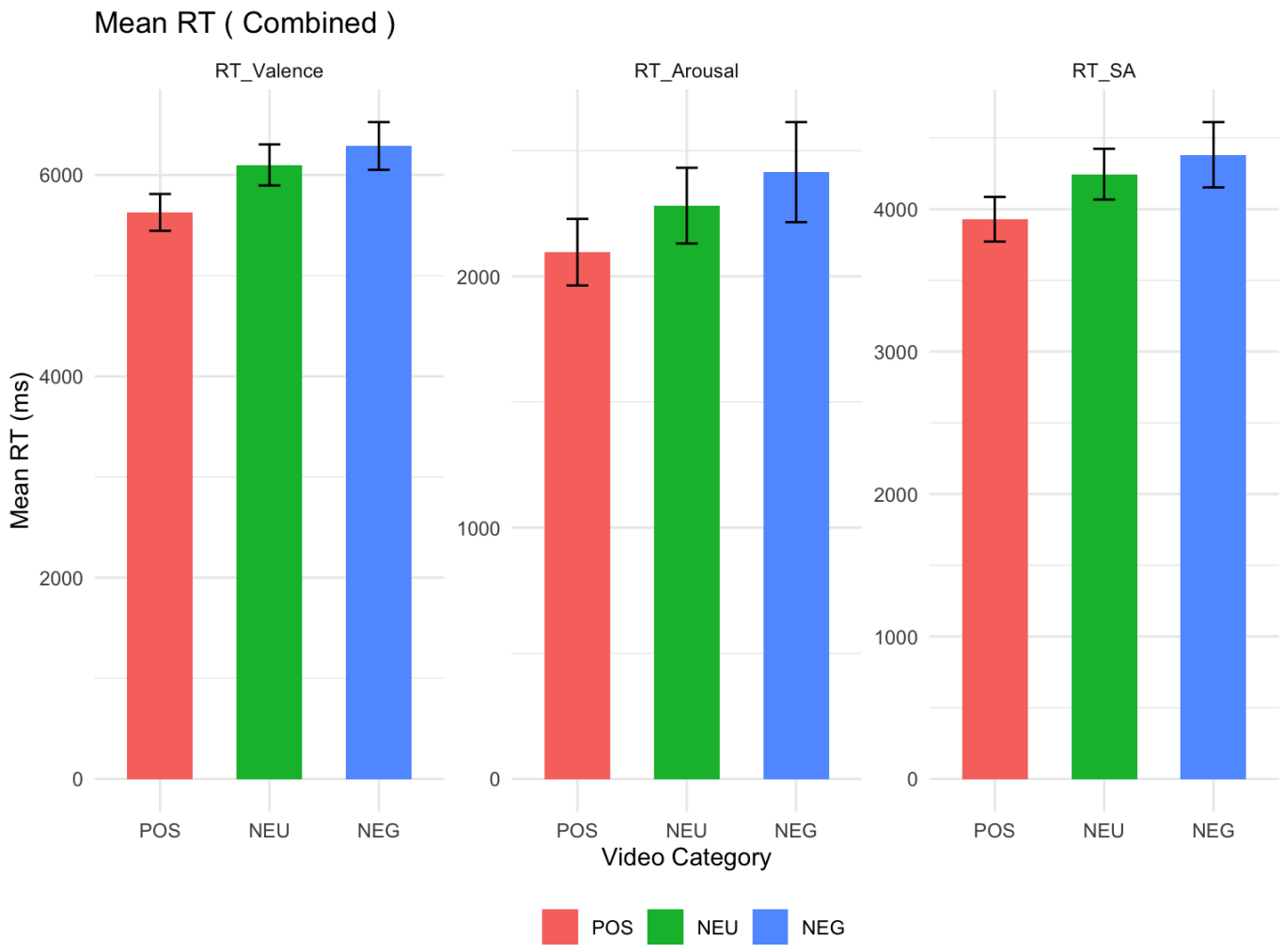
```
##
##
## ## 条形图（ Offline ）
```


Mean Ratings (Offline)
Valence
Arousal
SA
Mean Rating
0
2
4
6
POS
NEU
NEG
Video Category
POS
NEU
NEG


Mean RT (Offline)
RT_Valence
RT_Arousal
RT_SA
Mean RT (ms)
0
2000
4000
6000
500
1000
1500
3000
POS
NEU
NEG
Video Category
POS
NEU
NEG

```
##
##
## ## 条形图 ( Combined )
```


Mean Ratings (Combined)
Valence
Arousal
SA
Mean Rating
0
2
4
6
POS
NEU
NEG
Video Category
POS
NEU
NEG


Mean RT (Combined)
RT_Valence
RT_Arousal
RT_SA
Mean RT (ms)
0
1000
2000
3000
4000
6000
POS
NEU
NEG
Video Category
POS
NEU
NEG

2.2 SAQ regression

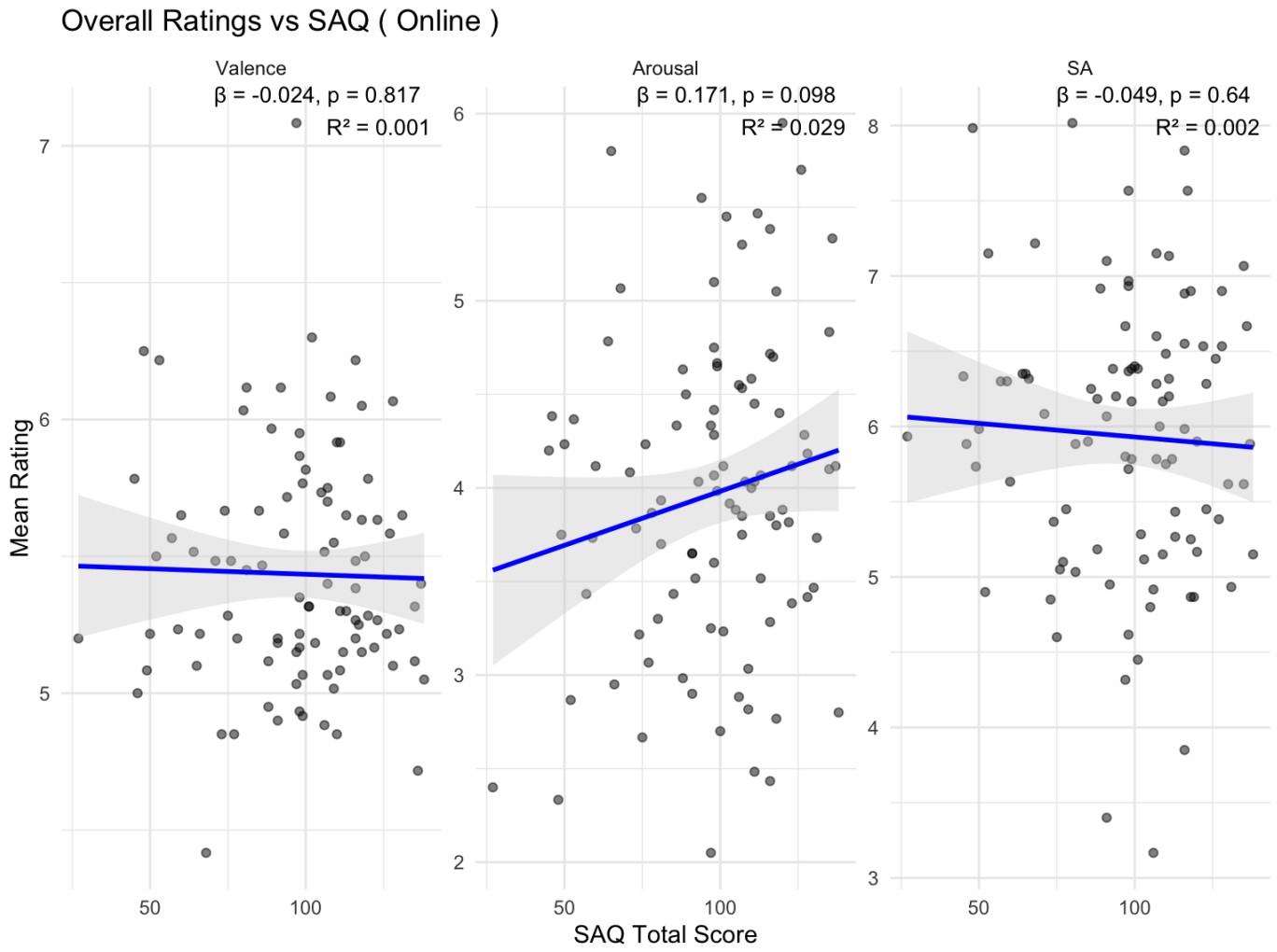
```
##
##
## ## SAQ 回归分析 ( Online )
```

```
## Warning: There was 1 warning in `summarise()`.
## ℹ In argument: `label = get_reg_label(cur_data(), "saq_sum", "score")`.
## ℹ In group 1: `dimension = Valence`.
## Caused by warning:
## ! `cur_data()` was deprecated in dplyr 1.1.0.
## ℹ Please use `pick()` instead.
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

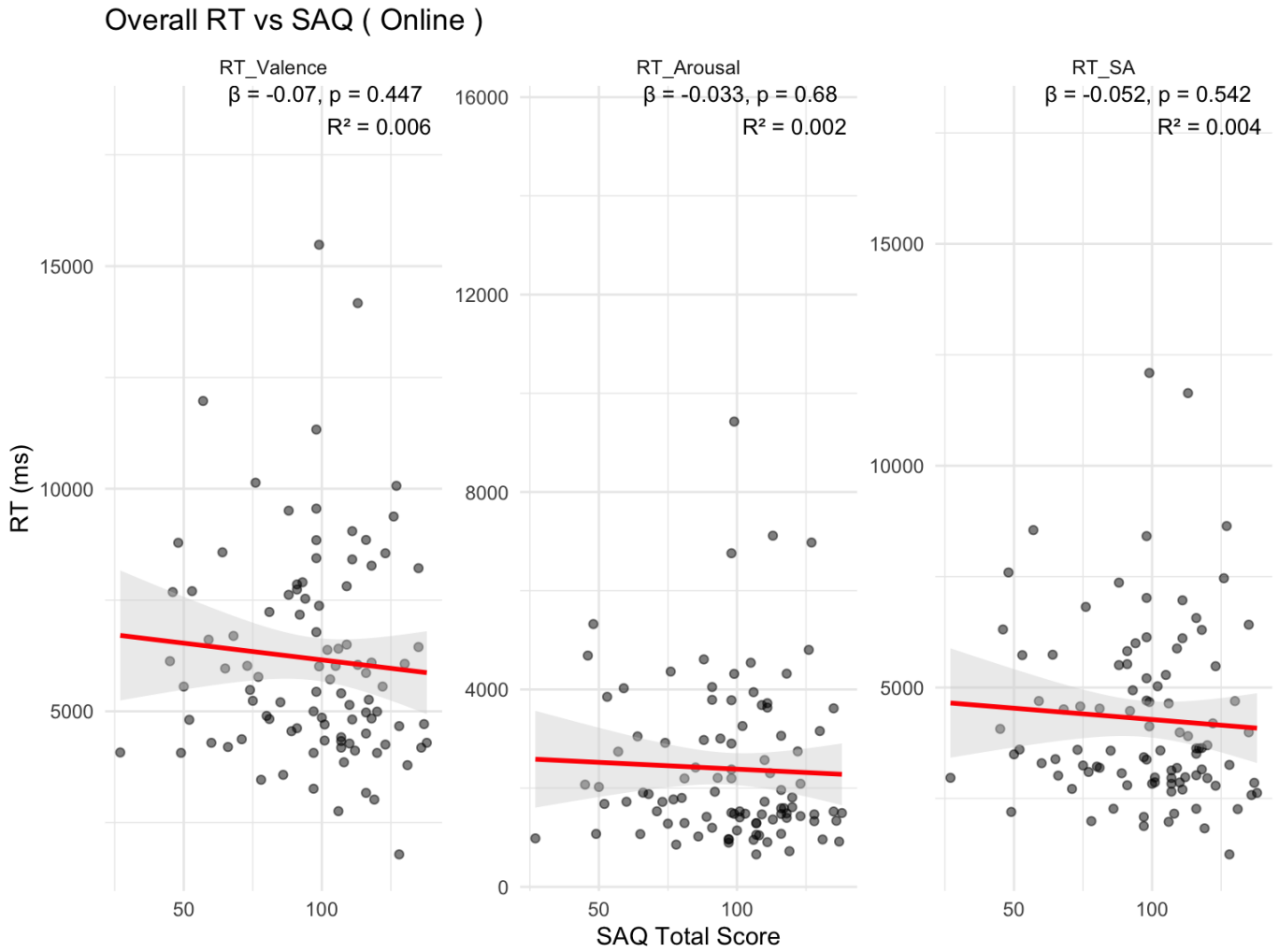
```
## Warning: Removed 9 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 9 rows containing missing values (`geom_point()`).
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

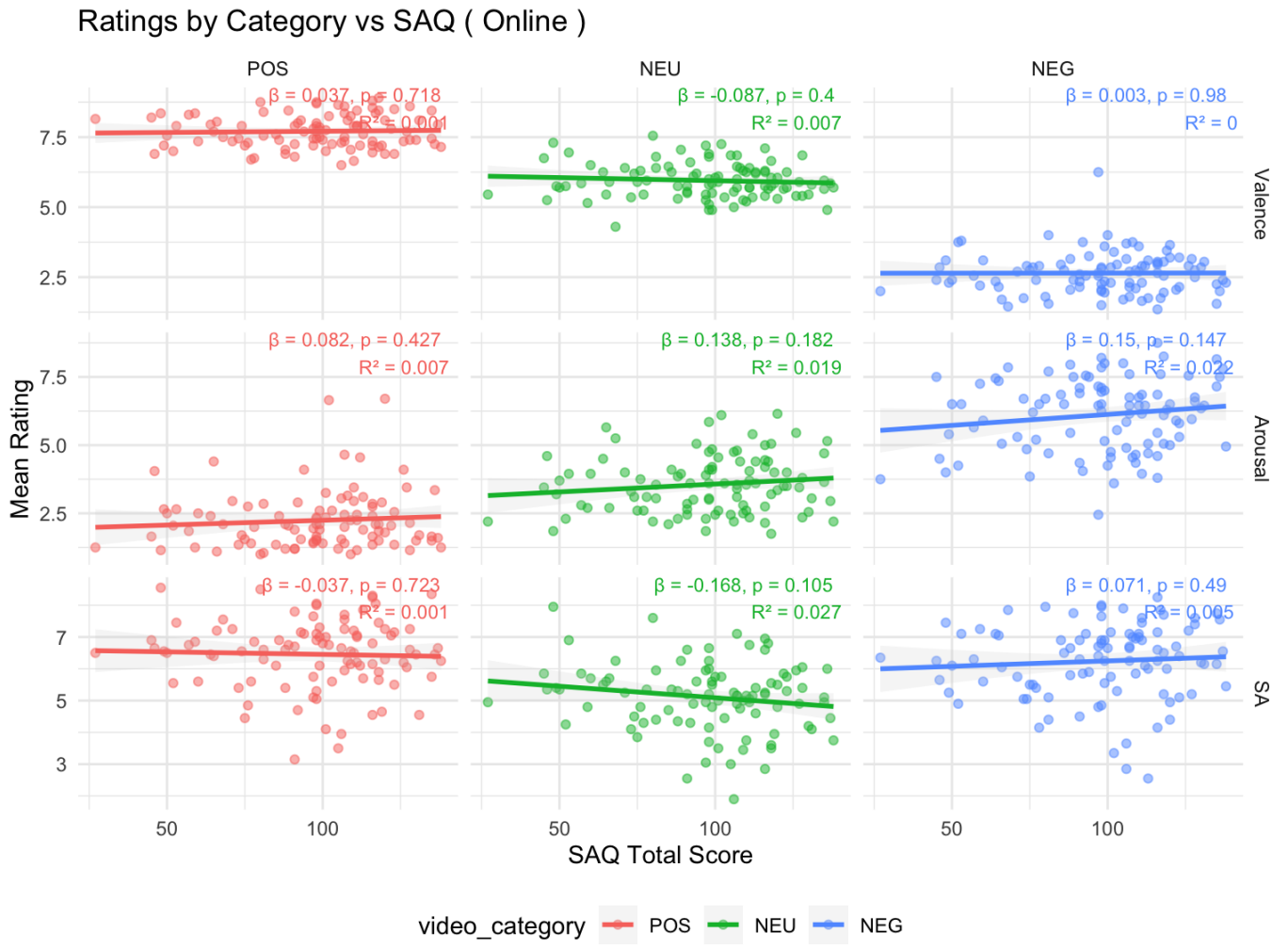
```
## Warning: Removed 9 rows containing non-finite values (`stat_smooth()`).
## Removed 9 rows containing missing values (`geom_point()`).
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

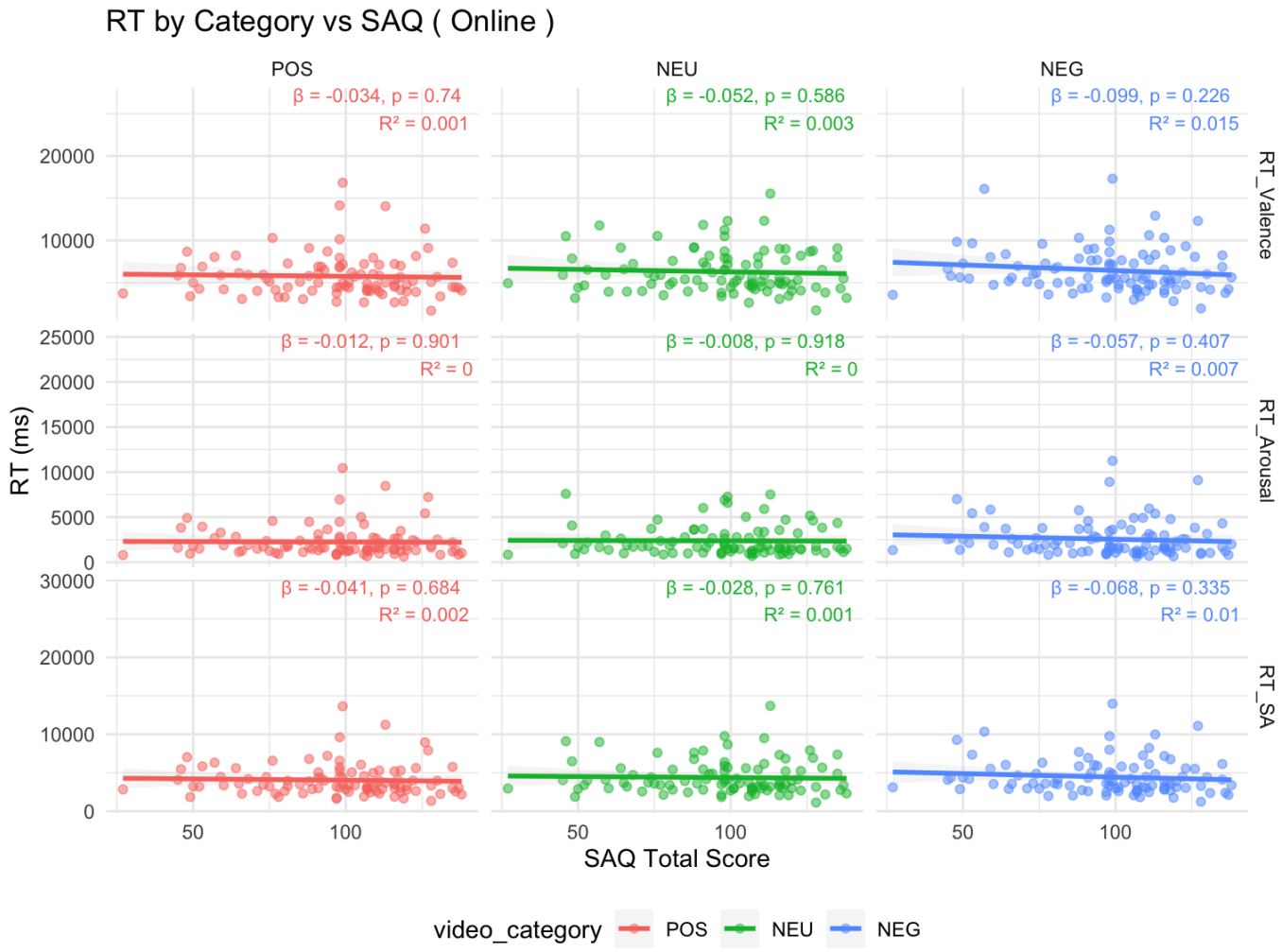
```
## Warning: Removed 27 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 27 rows containing missing values (`geom_point()`).
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 27 rows containing non-finite values (`stat_smooth()`).
## Removed 27 rows containing missing values (`geom_point()`).
```

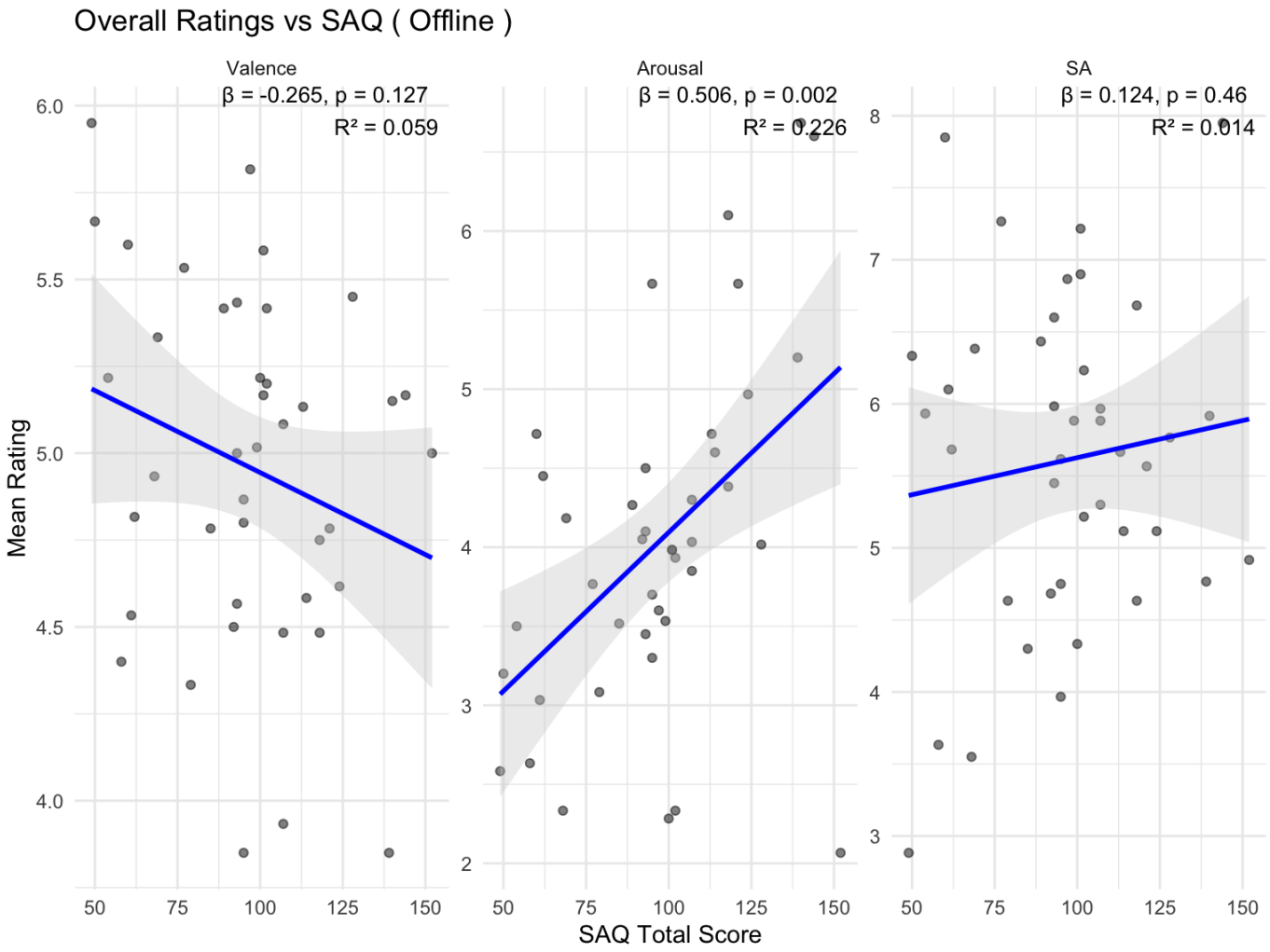



```
##
##
## ## SAQ 回归分析 ( Offline )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

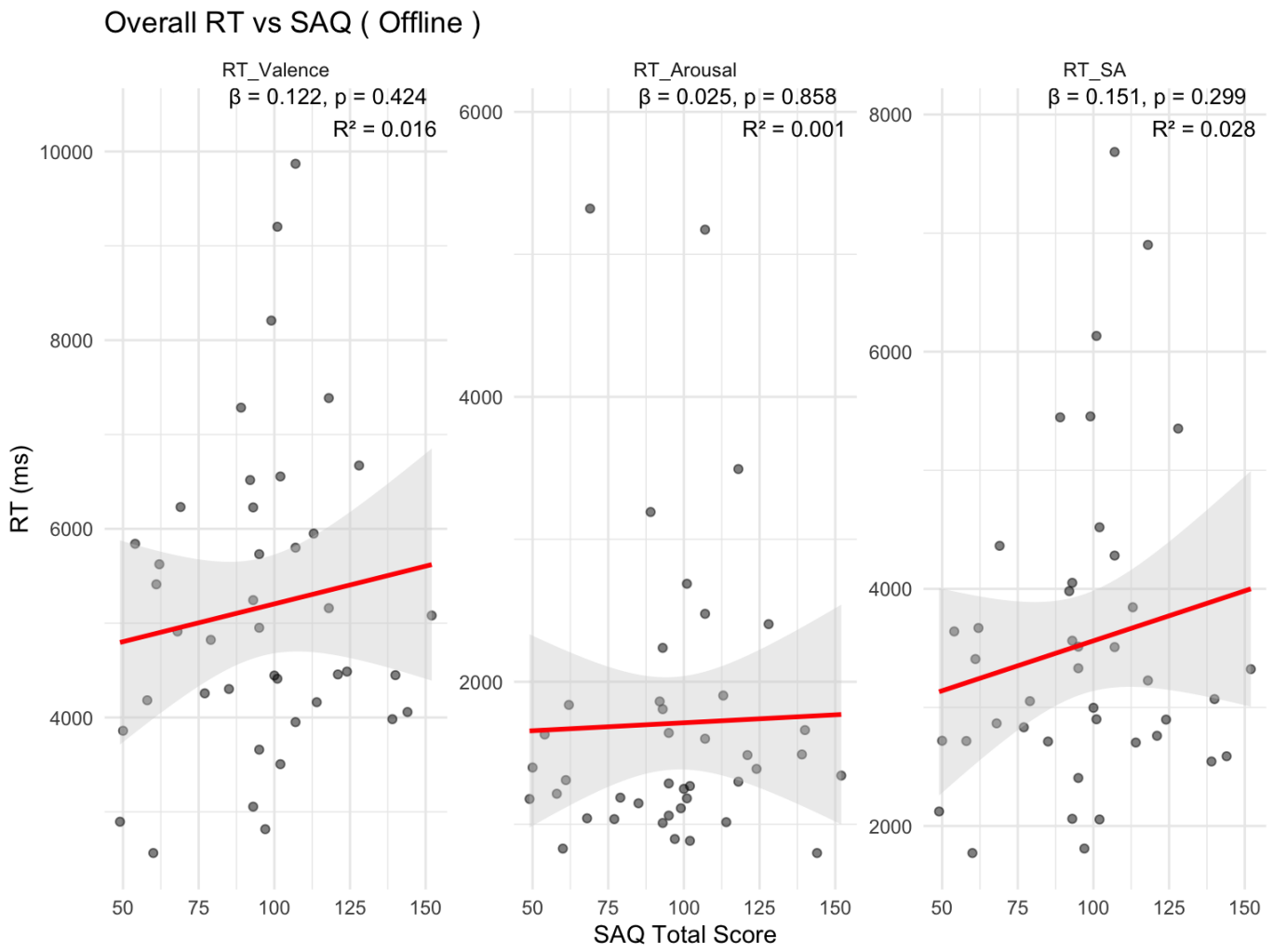
```
## Warning: Removed 33 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 33 rows containing missing values (`geom_point()`).
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

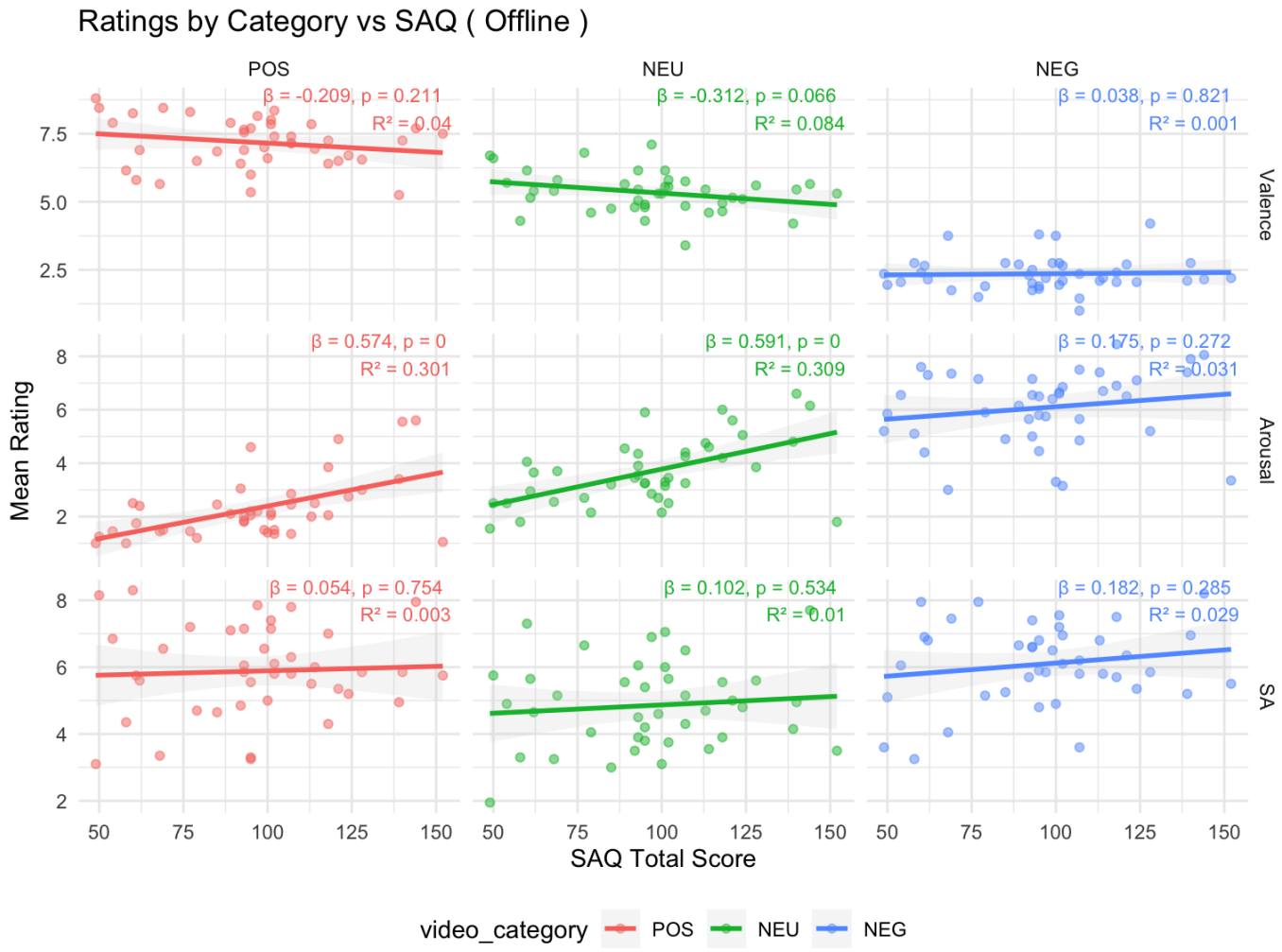
```
## Warning: Removed 33 rows containing non-finite values (`stat_smooth()`).
## Removed 33 rows containing missing values (`geom_point()`).
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

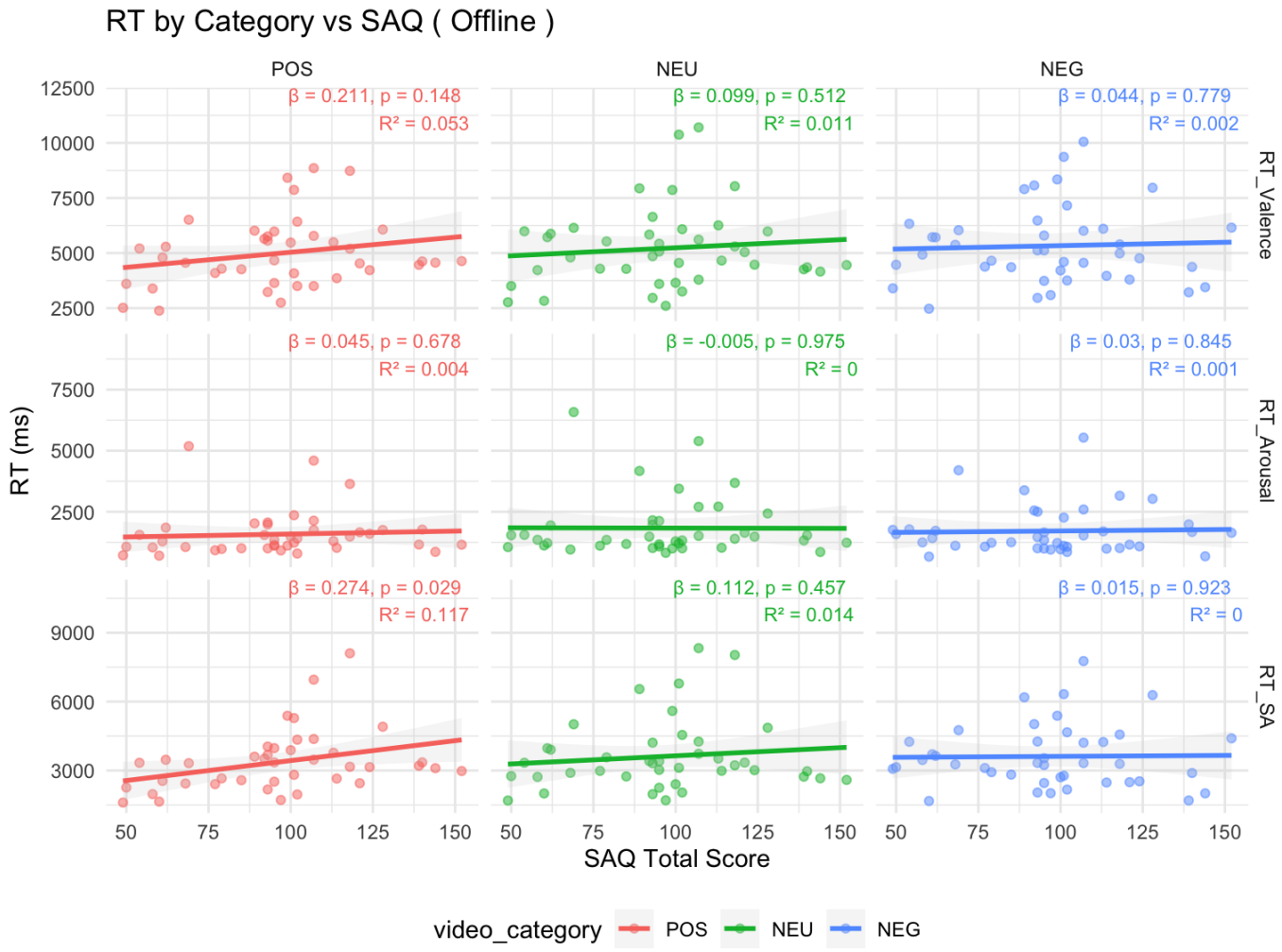
```
## Warning: Removed 99 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 99 rows containing missing values (`geom_point()`).
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 99 rows containing non-finite values (`stat_smooth()`).
## Removed 99 rows containing missing values (`geom_point()`).
```

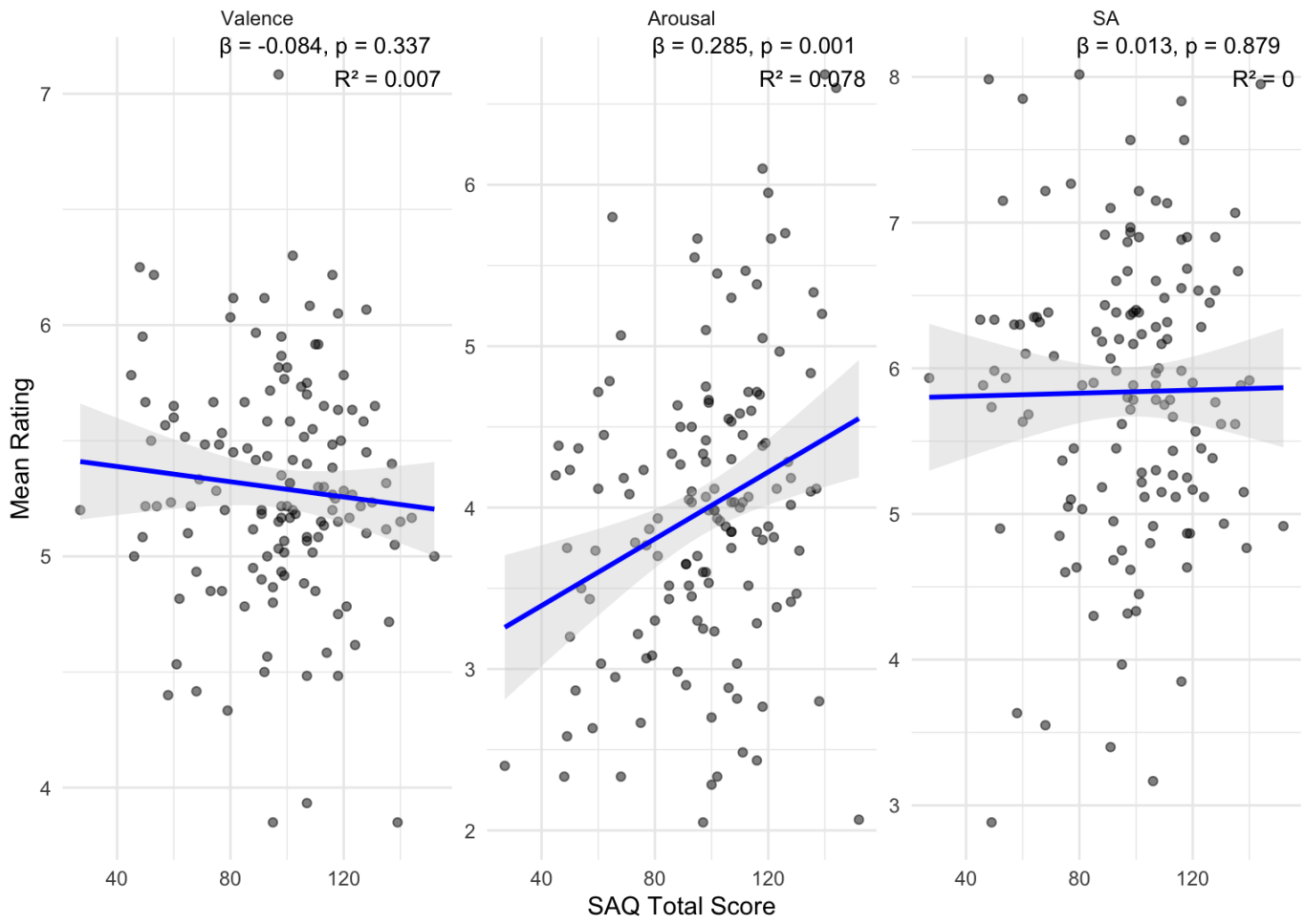



```
##
##
## ## SAQ 回归分析 ( Combined )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

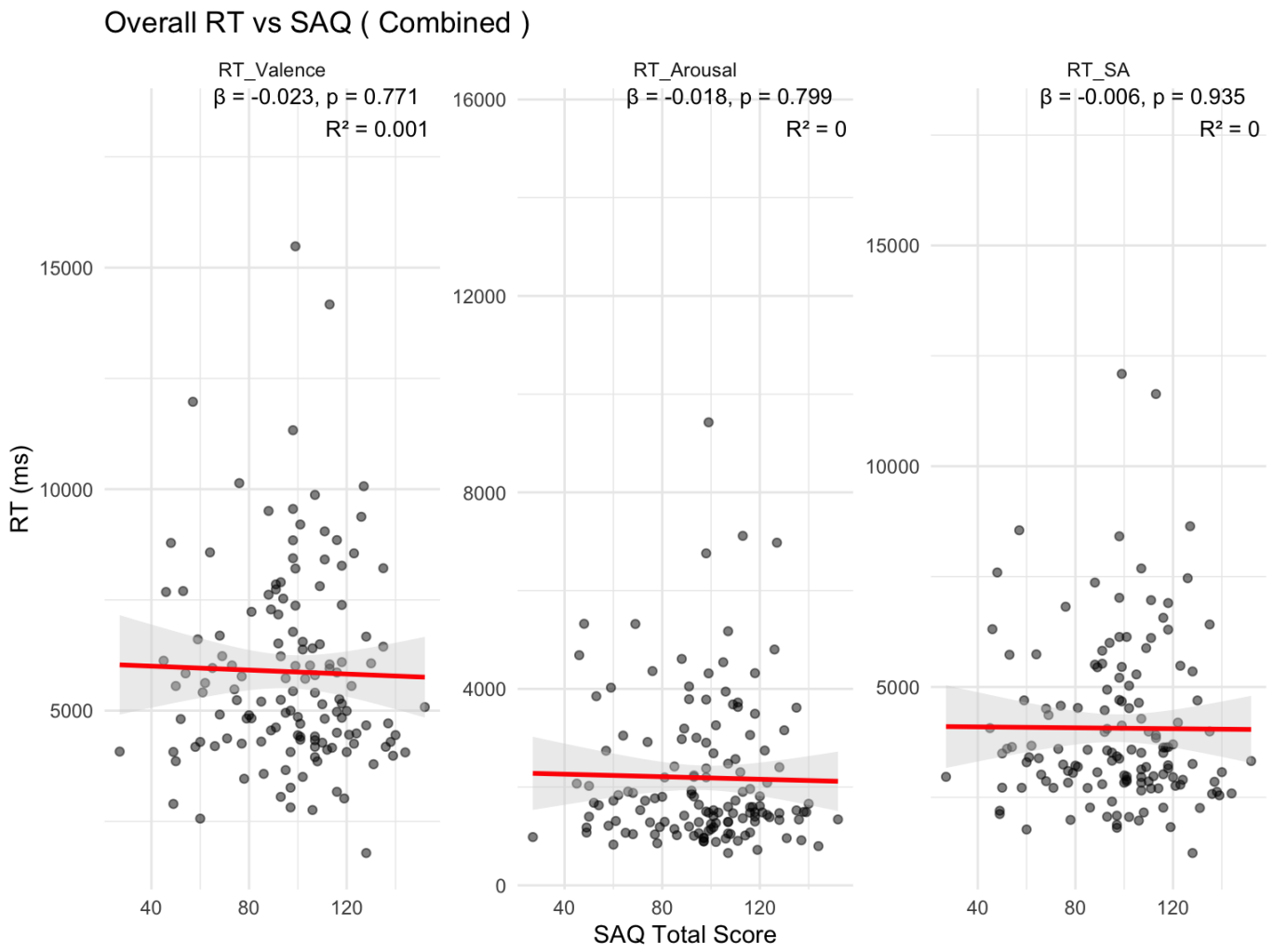
```
## Warning: Removed 42 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 42 rows containing missing values (`geom_point()`).
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

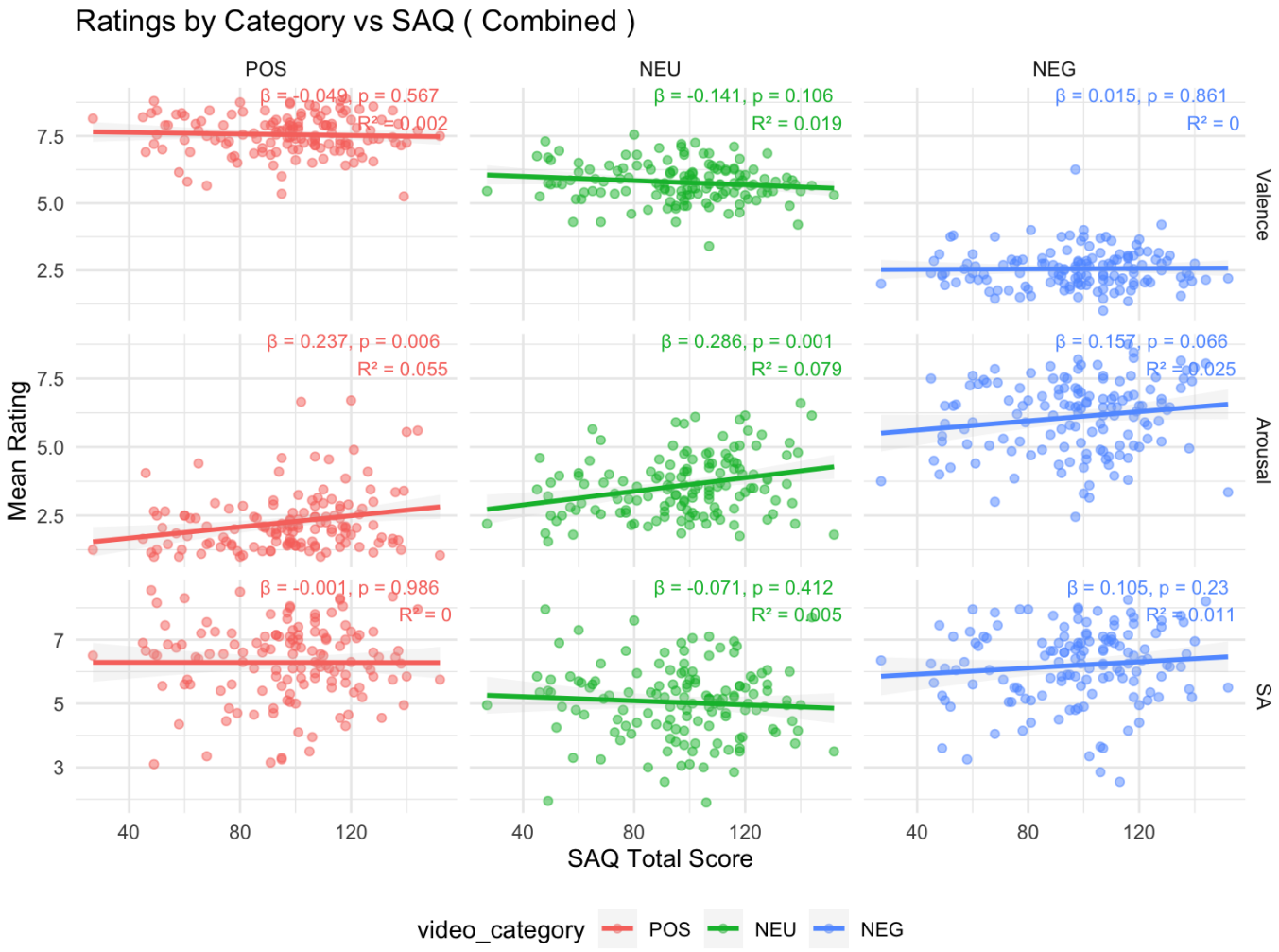
```
## Warning: Removed 42 rows containing non-finite values (`stat_smooth()`).
## Removed 42 rows containing missing values (`geom_point()`).
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

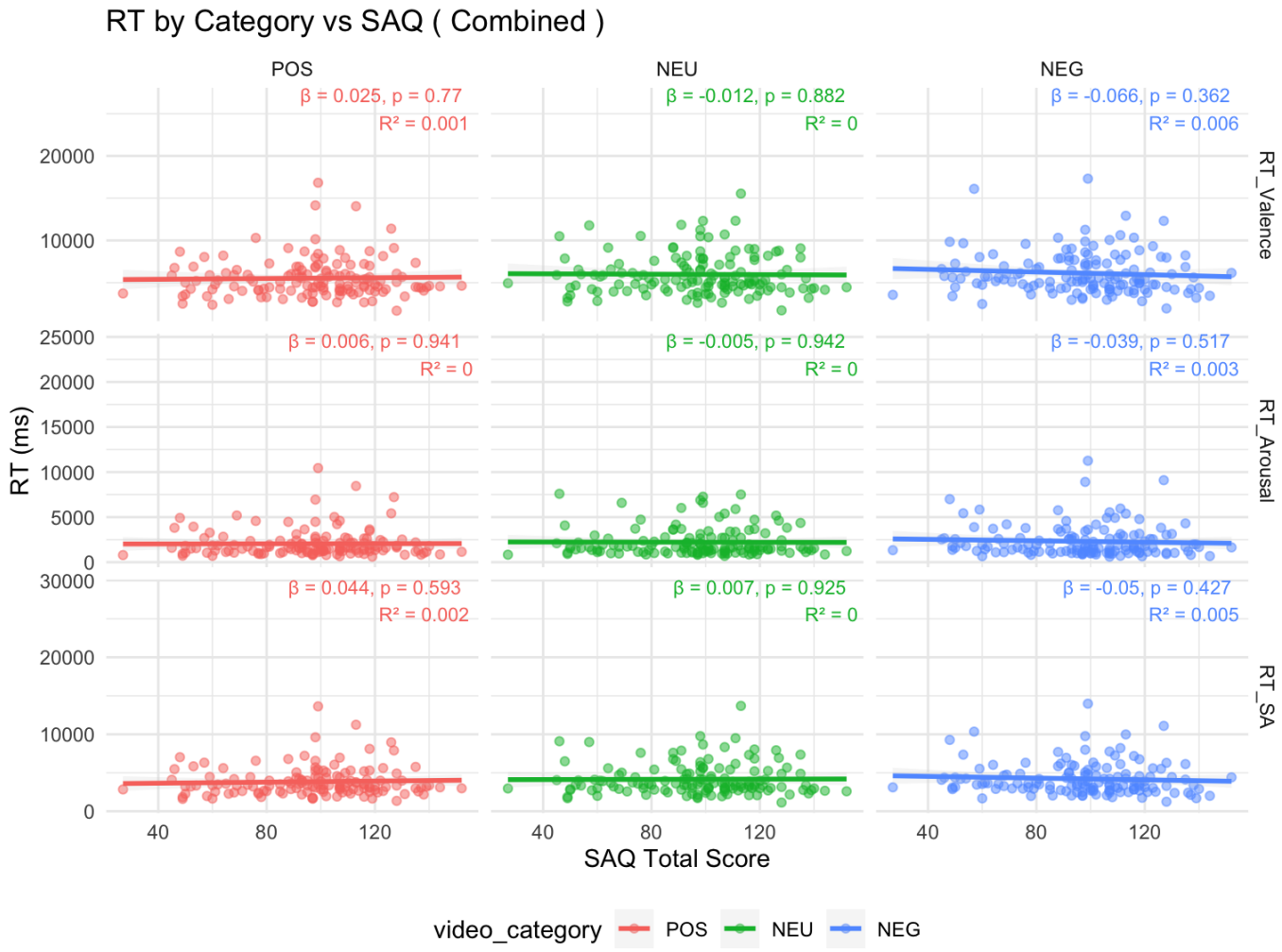
```
## Warning: Removed 126 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 126 rows containing missing values (`geom_point()`).
```



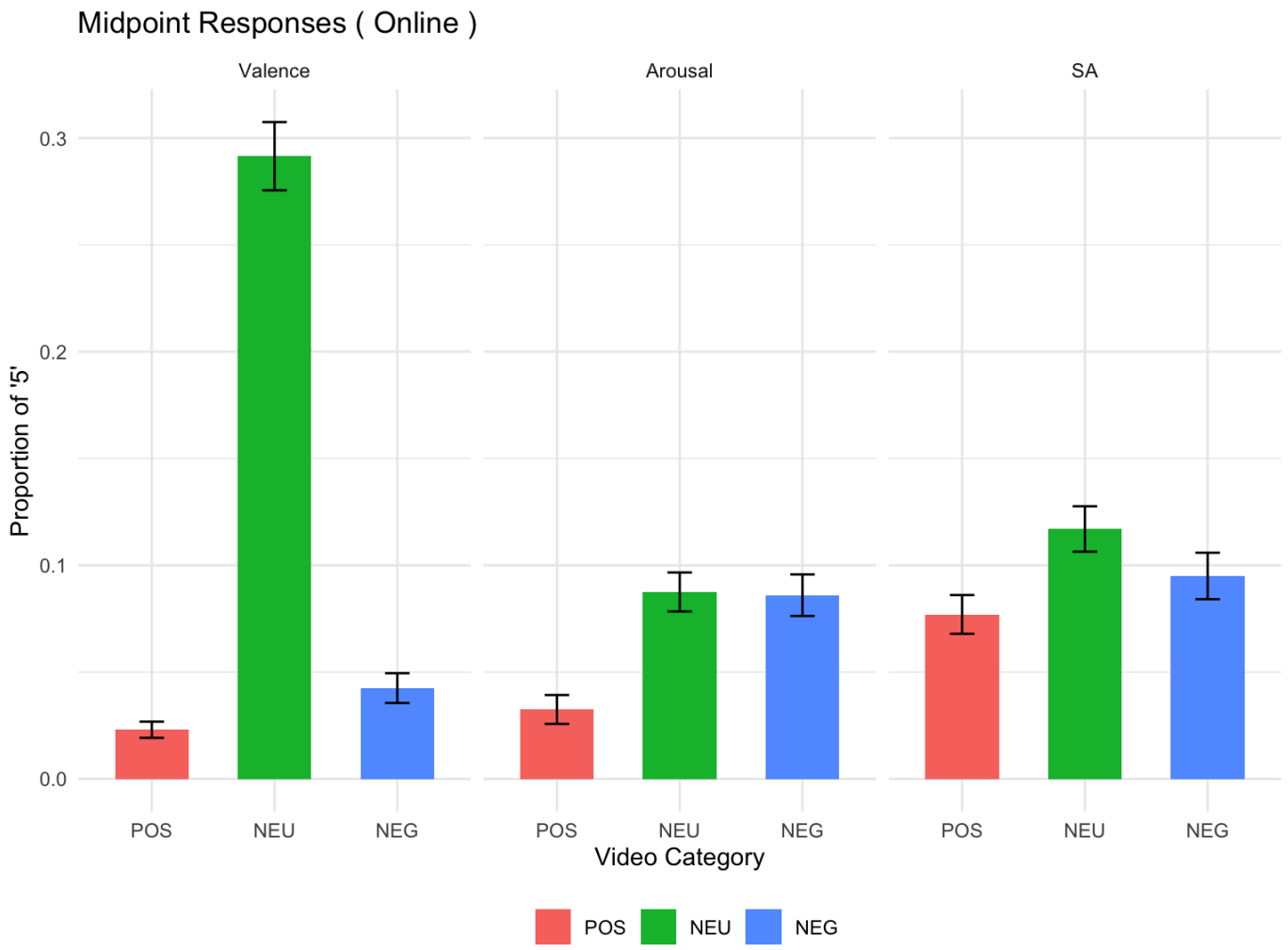

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 126 rows containing non-finite values (`stat_smooth()`).
## Removed 126 rows containing missing values (`geom_point()`).
```



2.3 mid point

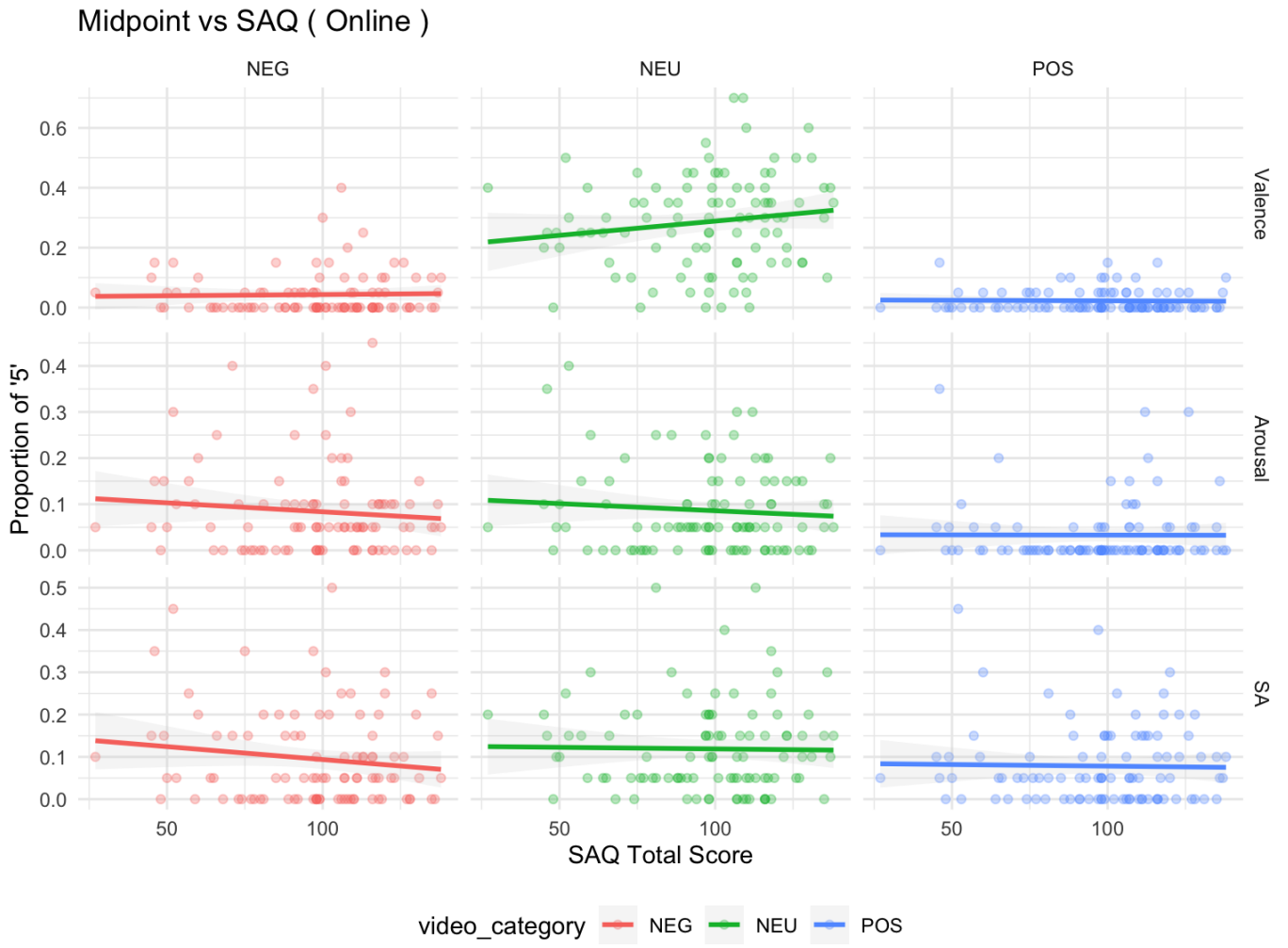
```
##
##
## ## 中点响应分析 （ Online ）
##
## ### ANOVA - ’5’ 比例
##
##
## ### LMM 固定效应（含 SAQ 协变量）
```



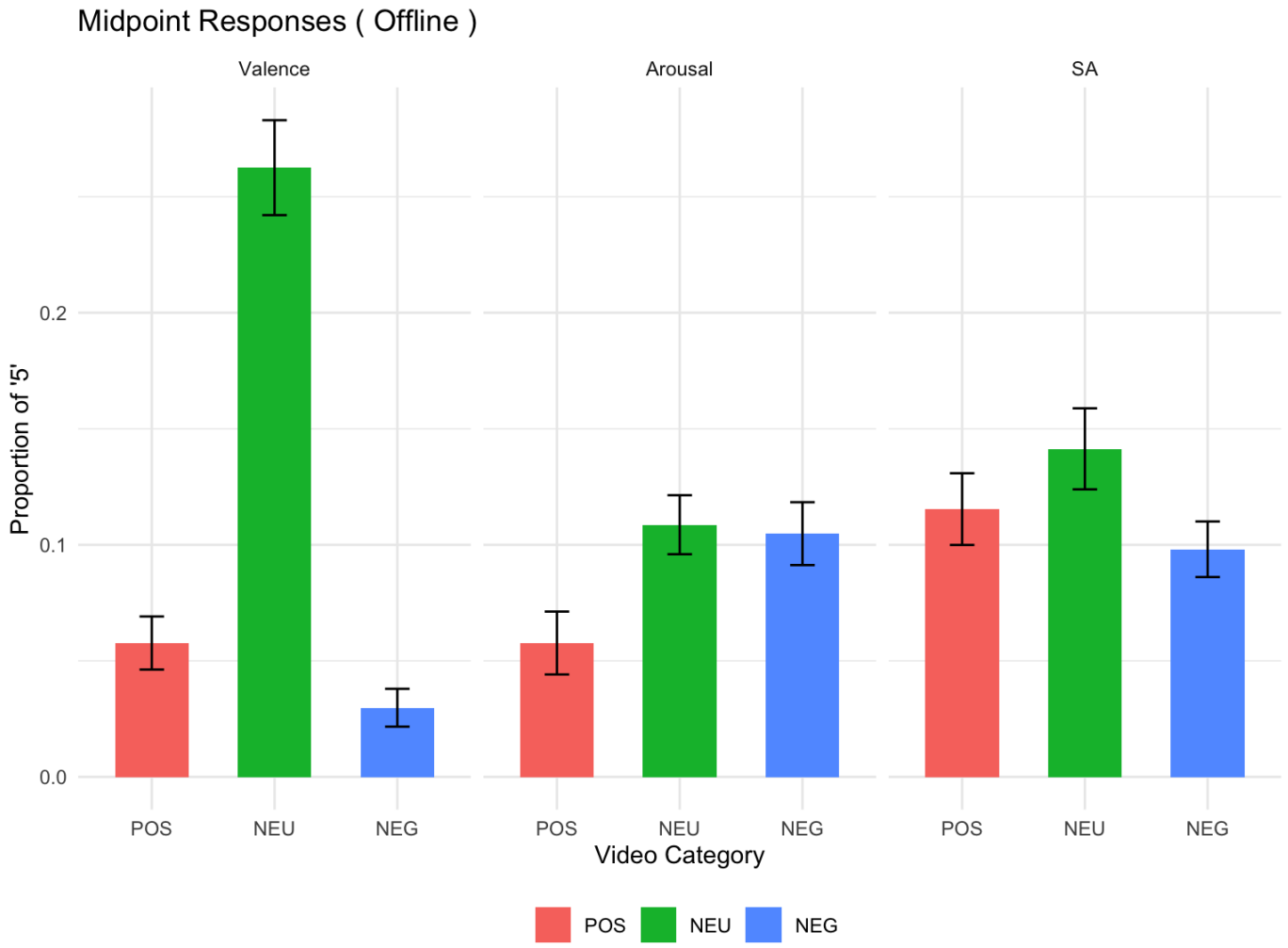

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 27 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 27 rows containing missing values (`geom_point()`).
```



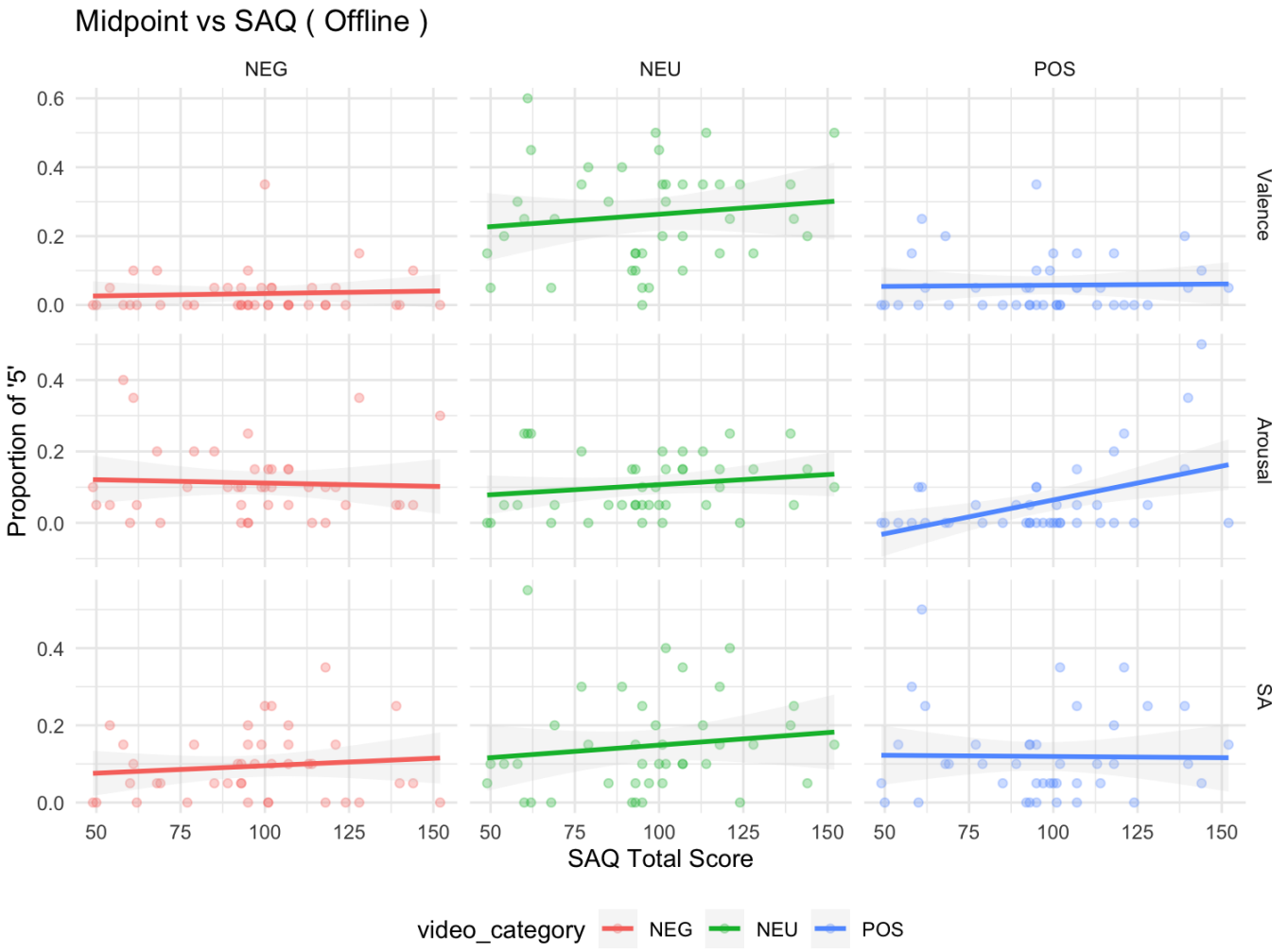

```
##
##
## ## 中点响应分析 （ Offline ）
##
## ### ANOVA - '5' 比例
##
##
## ### LMM 固定效应（含 SAQ 协变量）
```



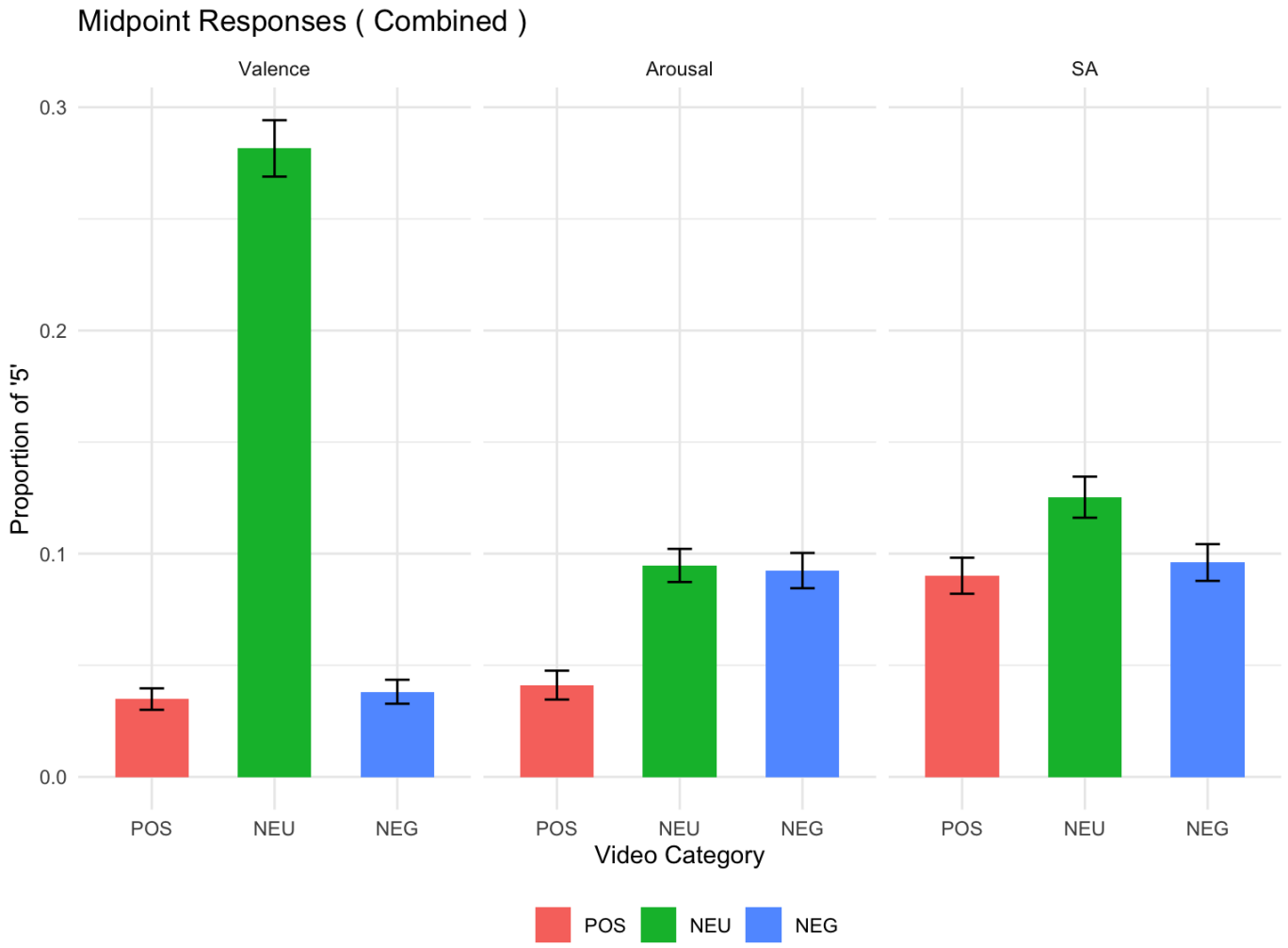

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 99 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 99 rows containing missing values (`geom_point()`).
```



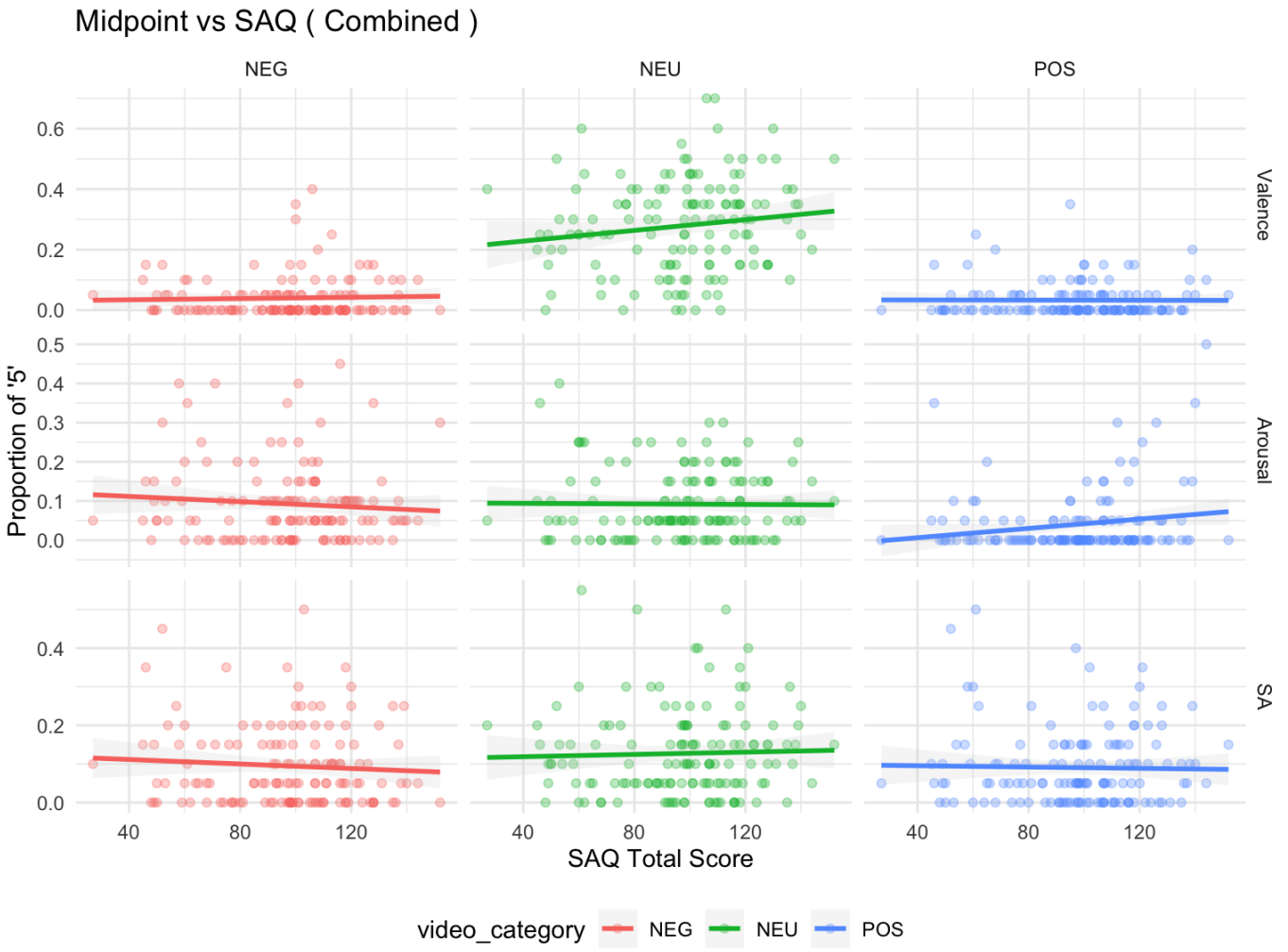

```
##
##
## ## 中点响应分析 （ Combined ）
##
## ### ANOVA - '5' 比例
##
##
## ### LMM 固定效应（含 SAQ 协变量）
```




```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 126 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 126 rows containing missing values (`geom_point()`).
```


Midpoint vs SAQ (Combined)
NEG
NEU
POS
Valence
Arousal
SA
Proportion of '5'
0.0
0.1
0.2
0.3
0.4
0.5
0.6
40
80
120
SAQ Total Score
video_category
NEG
NEU
POS